

6 Starch from plants is used in the production of food, paper and building materials.

The structural and chemical properties of starch depend on its amylose and amylopectin content.

(a) The table shows some of the bonds found in amylose and amylopectin.

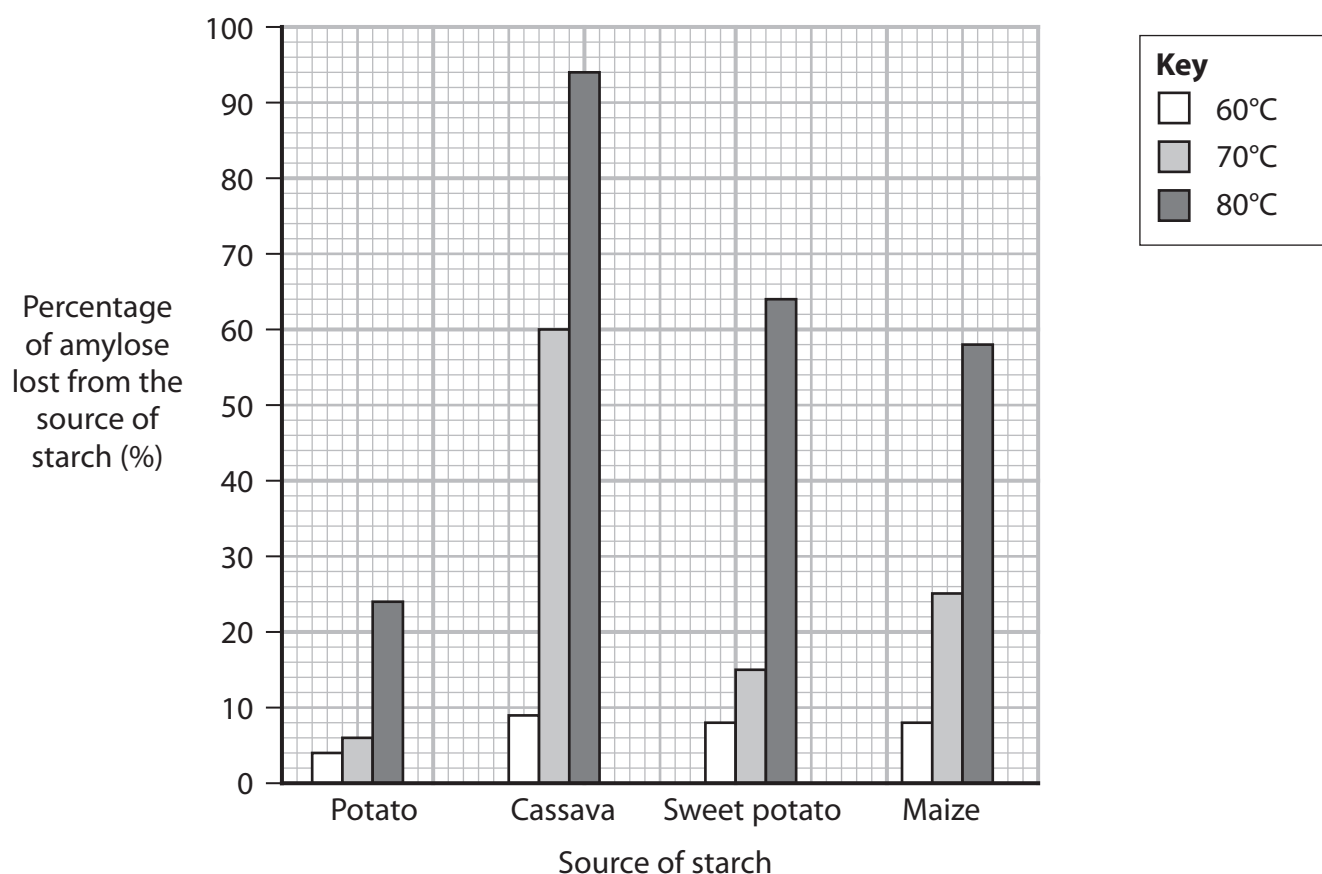
Put **one** cross ☐ in the appropriate box in each row to show whether the bond is present in these molecules.

(3)

Types of bond	Bonds found in			
	amylose only	amylopectin only	both amylose and amylopectin	neither amylose nor amylopectin
glycosidic bonds	☒	☒	☒	☒
1-6 α bonds	☒	☒	☒	☒
hydrogen bonds	☒	☒	☒	☒

(b) When sources of starch are heated in water, amylose moves into the water.

The graph shows the percentage of amylose that is lost from four sources of starch at three temperatures.



Describe **two** conclusions that can be made about the effects of temperature and source of starch on the loss of amylose.

(2)

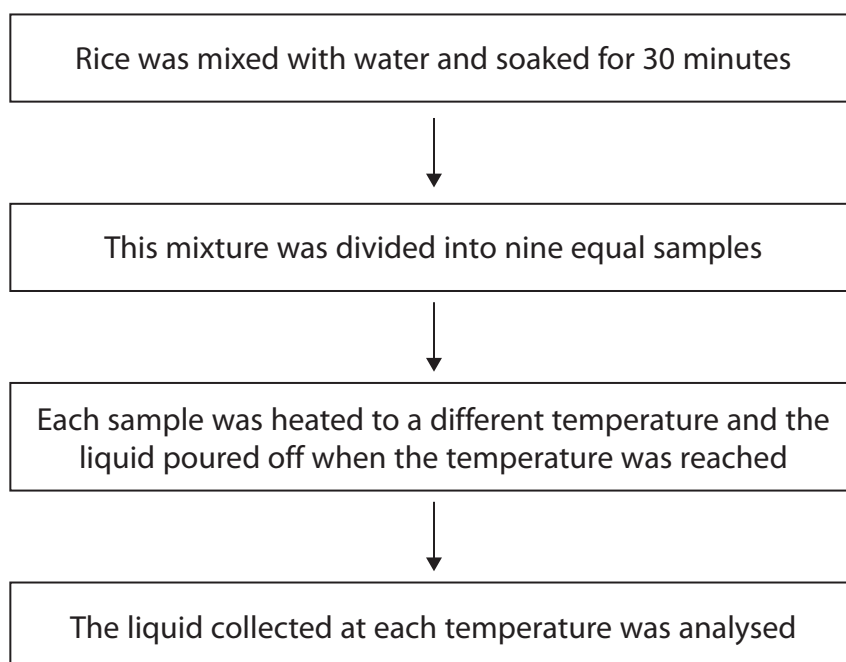
1 Effect of temperature

2 Effect of source of starch

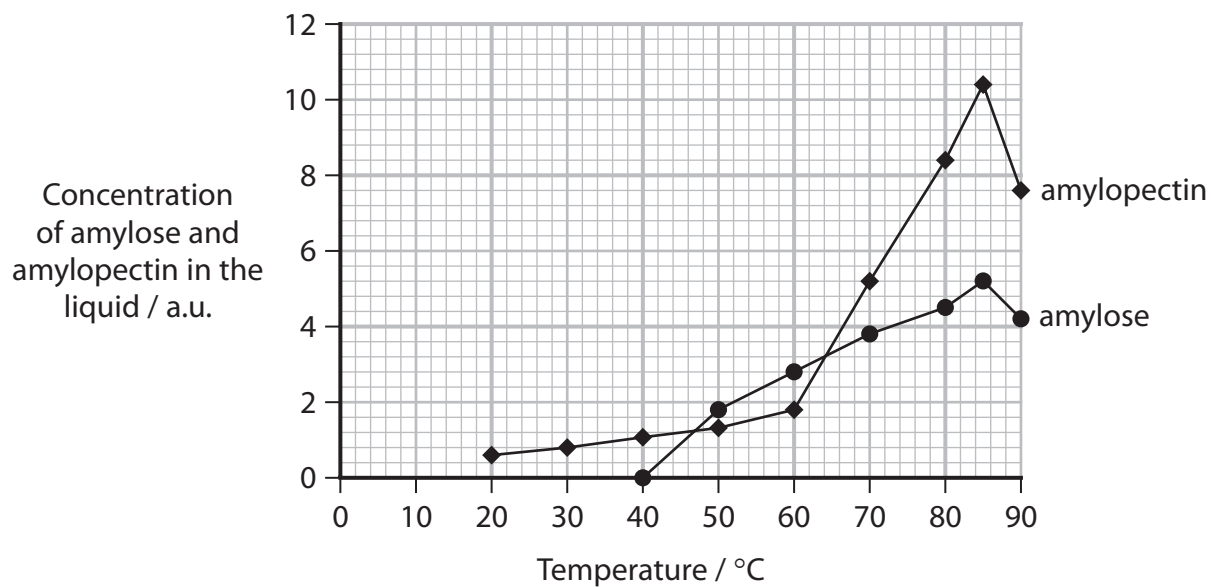
- (c) The loss of amylose and amylopectin during the cooking of rice affects the hardness and stickiness of the cooked rice.

The effect of temperature on the loss of amylose and amylopectin was investigated.

The diagram shows part of the method used.



The graph shows the results of this investigation.



- (i) Compare and contrast the effects of temperature on the loss of amylose and amylopectin from rice.

(3)

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- (ii) The investigation continued by adding a replacement liquid to finish cooking the rice.

The hardness and stickiness of the cooked rice was determined for each replacement liquid.

The table shows the results of this investigation.

Replacement liquid	Hardness of cooked rice / a.u.	Stickiness of cooked rice / a.u.
Water	19.42	1.19
Water containing amylose	27.63	1.21
Water containing amylopectin	20.43	1.71
Water containing both amylose and amylopectin	27.65	1.70

Comment on the effect of the replacement liquids on the hardness and stickiness of cooked rice.

Use the information in the table to support your answer.

(3)

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- (iii) Suggest why rice should be cooked in a measured volume of water, with the lid on, until all the cooking liquid has been absorbed.

Use the information in the table to support your answer.

(2)

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(Total for Question 6 = 13 marks)

